

“

Seasonal Inventory Management

ToyJoy Seasonal Inventory Management

”

EXECUTIVE SUMMARY

- This analysis evaluates ToyJoy inventory performance using 3 years of data.
- Key issues identified include excess inventory, frequent stockouts, and inefficient replenishment strategies.
- Recommendations focus on demand-based inventory planning, reorder optimization, and seasonal adjustments to improve efficiency and profitability.

Business Objective & Goal

Objective

- ▶ Optimize inventory using historical data
- ▶ Reduce overstock & holding costs
- ▶ Prevent stockouts & lost sales
- ▶ Analyze seasonal demand trends
- ▶ Improve stock levels & reorder planning

Goal

- ▶ Minimize overstock
- ▶ Reduce stockouts
- ▶ Maximize sales & profitability

Dataset

Overview

- ▶ The dataset contains transactional and inventory data for ToyJoy, covering multiple products, stores, and seasonal periods.
- ▶ **Time Period:** 3 years of historical data (weekly level)
- ▶ **Business Scope:** Retail + Online toy sales

Data Cleaning & Preparation

- ▶ Checked for missing values and handled appropriately
- ▶ Ensured consistency in category names and formats
- ▶ Verified numerical fields (UnitsSold, UnitsInStock, Price)
- ▶ **Assumptions**
- ▶ Lead time and safety stock were assumed due to lack of data
- ▶ Stockout threshold was defined as ≤ 25 units

Exploratory Data Analysis (EDA)

Units Sold

50461

Stockout Rate

3.39%

Excess Inventory

2564108.37

**Inventory Turnover
Rate/Product**

Action Figure	115.8
---------------	-------

Board Game	111.9
------------	-------

Doll	112.9
------	-------

Puzzle Set	120.8
------------	-------

Seasonal Demand

Back to School	9868
----------------	------

Christmas	9911
-----------	------

Diwali	9686
--------	------

Normal days	10459
-------------	-------

Summer Holiday	10537
----------------	-------

**Revenue Contribution
per Product**

Action Figure	25.06%
---------------	--------

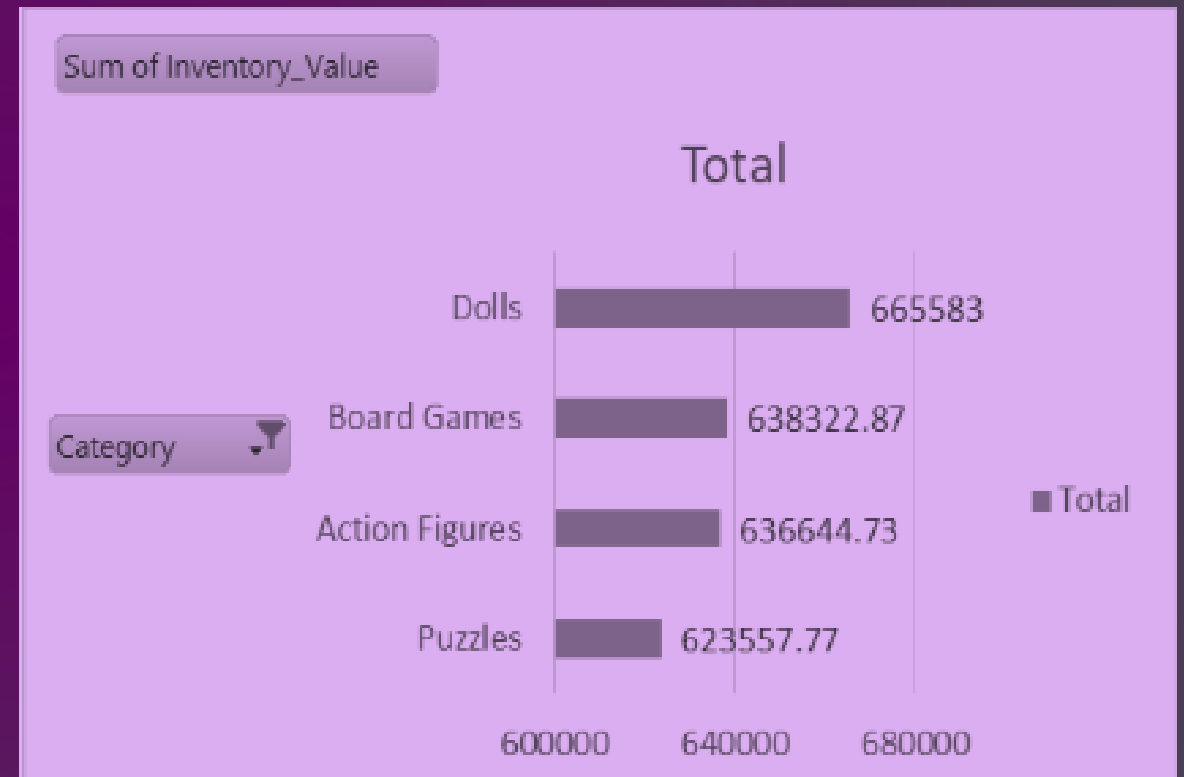
Board Game	24.01%
------------	--------

Doll	25.54%
------	--------

Puzzle Set	25.39%
------------	--------

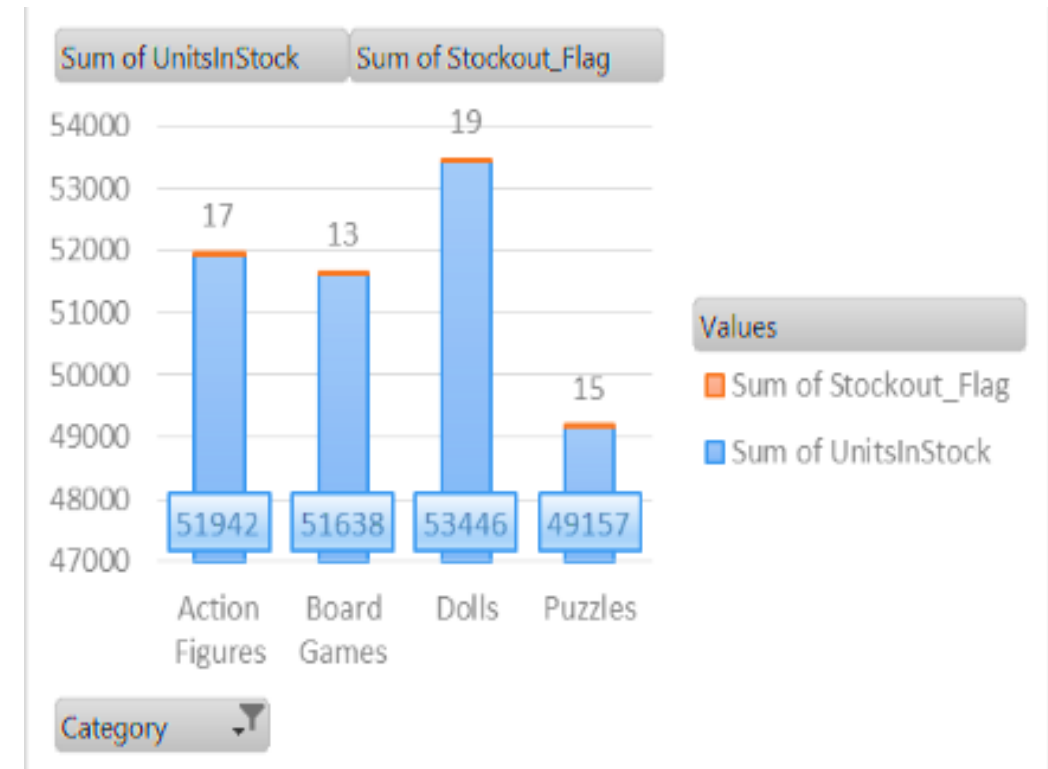
Excess Inventory & Capital Blockage by Product Category

1. **Dolls hold the highest inventory value (665K),** indicating excess stock and capital blockage
2. **Board Games and Action Figures also show high inventory levels,** suggesting overstock risk
3. Inventory is **not aligned with actual demand,** leading to inefficient utilization
4. **Opportunity:** Optimize stock levels using demand-based planning to improve cash flow and efficiency



Products frequently out of stock

- ▶ The analysis of stock availability using a threshold of **25 units**
- ▶ **Dolls** have the highest stockout occurrences (**19 instances**), indicating frequent low inventory levels and high risk of lost sales.
- ▶ **Action Figures** follow with **17 instances**, also showing significant stock availability issues.
- ▶ **Puzzles** recorded **15 stockouts**, indicating moderate stock shortages.
- ▶ **Board Games** have the lowest stockout count (**13 instances**), suggesting relatively better inventory management.



Insights

Seasonal Sales Trends & Demand

- ▶ **Summer Holiday shows the highest sales (10,537 units)** → Peak demand period
- ▶ **Normal Days also maintain strong sales (10,459 units)** → Stable year-round demand
- ▶ **Christmas and Back to School show moderate performance**
- ▶ **Diwali records the lowest sales (9,686 units)** → Lower seasonal demand
- ▶ Overall, sales are **fairly stable with slight seasonal variation**, with demand increasing during school holidays



Recommendations

Inventory Adjustment Recommendations

High-Demand Products (Increase Stock)

•**Products:** Dolls, Puzzles

Why:

- High sales (strong demand)
- High inventory turnover
- Frequent stockouts

Action:

- Increase inventory levels
- Maintain safety stock
- Replenish frequently

Underperforming Products (Reduce Stock)

•**Product:** Board Games

Why:

- Lower sales
- Low inventory turnover
- Fewer stockouts

Action:

- Reduce stock levels
- Avoid over-ordering
- Monitor demand before restocking

Reorder Optimization Strategies

- Set demand-based reorder points (Like as: Dolls ~117 units, Puzzles ~115 units)
- Trigger replenishment at reorder levels to prevent stockouts
- Maintain safety stock for high-demand products
- Adjust order quantities based on demand (higher for fast-moving, lower for slow-moving)
- Align inventory with seasonal demand (increase stock before peak periods)
- Monitor key KPIs (stockouts, inventory turnover, inventory value)
- Use a data-driven approach to reduce overstock and improve cash flow

Business Impact & Final Conclusion

Impact

- ▶ Reducing stockouts can increase revenue by improving product availability
- ▶ Optimizing inventory can reduce holding costs and improve cash flow
- ▶ Better demand planning leads to efficient warehouse utilization
- ▶ Improved inventory decisions can enhance customer satisfaction and service levels

Conclusion

- ▶ Inventory inefficiencies are driven by poor demand alignment and replenishment timing
- ▶ High-demand products like Dolls need better stock planning, while excess inventory impacts cash flow
- ▶ Data-driven inventory strategies can reduce waste, improve availability, and increase profitability and efficiency

Dashboard

Seasonal Inventory Management

Total Units Sold
50,461

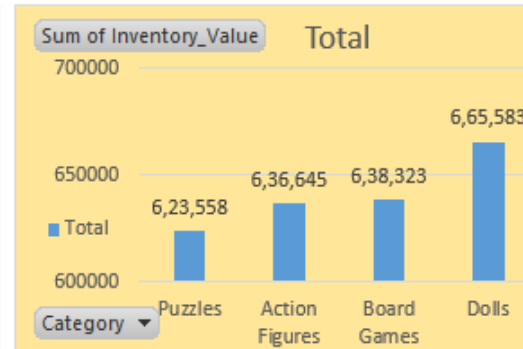
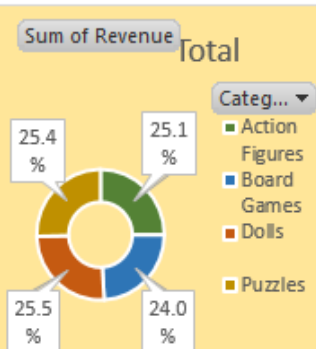
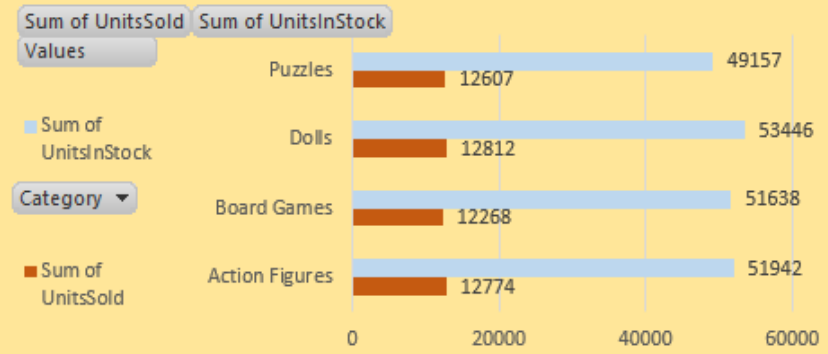
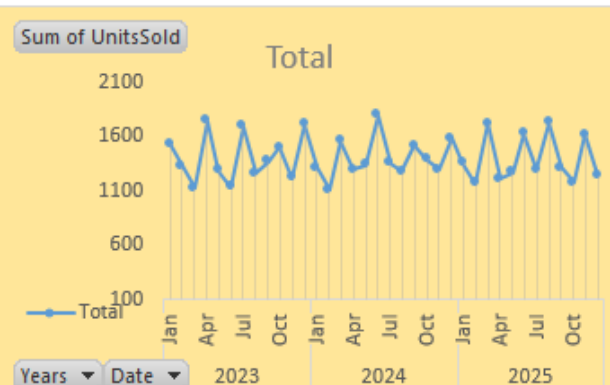
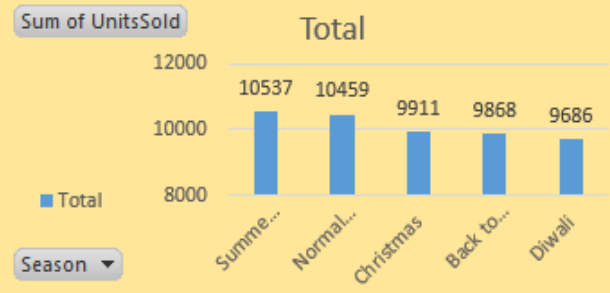
InventoryTurnoverRate/Product:

Action Figure A: 115.8
Board Game D: 111.9
Doll C: 112.9
Puzzle Set B: 120.8

Stockout Rate: 3.39%

Total Inventory Value
2,564,108

Avg Inventory Turnover
~115



StoreID

101

Category

Action Figures

Board Games

Dolls

Puzzles

Season

Back to School

Christmas

Diwali

Normal days

Summer Holiday